Case Study Plan: Managing a Software Database for an Educational Institution

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Introduction

Gulf Coast State University (GCSU), a fictitious public university with approximately 30,000 students, uses hundreds of software titles across 12 colleges, the library, and administrative offices. Currently, individual departments track licenses in separate spreadsheets, making it difficult to answer key questions during vendor audits, such as which versions are installed, on which computers, under which license terms, and when licenses expire. The Chief Information Officer (CIO) has appointed me as project leader to oversee a team developing a centralized software-tracking database for the Office of Information Technology (IT). This database will record software titles, developers or vendors, versions, license types and terms, departments using each title, computers where software is installed, and installation dates.

This plan follows the six-phase model described by Baars (2006): initiation, definition, design, development, implementation, and follow-up. The model divides the work into smaller, manageable elements and ends each phase with a decision point to review progress before moving on (Baars, 2006). Throughout the project, I will manage the five control factors the handbook identifies: time, money, quality, organization, and information. The project is planned for approximately 26 weeks.

Project Implementation Plan

Phase 1: Initiation

The initiation phase explores the idea, tests feasibility, identifies who should be involved, and confirms sufficient support (Baars, 2006). I will interview the CIO (the client and sponsor), the manager of IT Asset Management (the customer), and representatives from Procurement and

the Office of General Counsel to confirm why the database is needed: to reduce the danger of costly license violations and to cut the time spent preparing for audits. I will then write a project proposal using the handbook's sample project plan (Baars, 2006, Appendix 10), including a SMART goal, a global budget for the whole project, and a detailed budget for the definition phase.

Two agreements are critical at this stage. First, the handbook notes that conflicts often arise when there is uncertainty about whether a project will deliver a prototype or a fully operational product (Baars, 2006). This project will deliver a *working product*, including maintenance, instructions, and operational management. Second, I will establish clear boundaries. The database will track software and license information and link it to computers, but it will not replace the hardware inventory system, automate software purchasing, or scan the network for installed programs. Declining such additions helps maintain the project schedule, as expanding scope often prevents projects from meeting their original objectives (Baars, 2006).

- **Responsible:** Project leader (me); CIO approval.
- **Stakeholders:** CIO, IT Asset Management, Procurement, General Counsel, Information Security.
- **Deliverables:** Approved and funded project plan, global budget, initial risk log (Baars, 2006, Appendix 8).
- **Decision point:** Are we going to do this project?

Phase 2: Definition

In the definition phase, the team identifies what every party expects from the result and records those requirements as clearly as possible (Baars, 2006). I will hold workshops with the

business analyst, license administrators, desktop-support technicians, department IT liaisons from at least four colleges, Procurement, and Information Security. The handbook stresses including end users, not just the paying client, because users will actually work with the result (Baars, 2006). Here, the end users are the IT technicians who will enter installations and the license administrators who will run compliance reports.

Requirements will be organized using the four categories Wijnen et al. (2004, as cited in Baars, 2006) suggest. *Preconditions* include vendor license terms, state records-retention rules, and university security policy. *Functional requirements* include storing seat counts and expiration dates and producing a report of licenses expiring within 90 days. *Operational requirements* include, for example, reducing audit preparation time by at least 50% and supporting 30 simultaneous IT users. *Design limitations* consist of building on the university's existing database platform and using campus single sign-on. Learning from the handbook's example of a web application built for the wrong browser, the list will also state which browsers and operating systems must be supported (Baars, 2006). The team will negotiate and resolve conflicting requirements, such as departments wanting open access versus Security wanting restricted access, before approval.

- **Responsible:** Project leader and business analyst.
- **Stakeholders:** IT Asset Management, technicians, department IT liaisons, Procurement, Information Security, system administrator.
- **Deliverables:** Approved list of requirements, a written list of what will *not* be built, and an updated control-factor report.

- **Decision point:** Which requirements should the result fulfill, and is it possible within the budget?

Phase 3: Design

The design phase uses the requirements list to decide how the result will look and work (Baars, 2006). The database developer will create a conceptual data model showing how software titles, vendors, versions, licenses, departments, computers, and installations relate to one another. The interface designer will prepare screen mock-ups for data entry, search, and the main reports. Following the handbook's advice for IT projects, we will build a quick "dummy" (a partly working version) and show it to technicians and license administrators, because a dummy resembles the finished product more closely than paper drawings (Baars, 2006).

The handbook describes a project that had to be redone because the department head responsible for approving the designs never actually reviewed them (Baars, 2006). To prevent this, the IT Asset Management manager will sign off on the design document in writing, and the action-and-decision list will record the approval (Baars, 2006, Appendix 6). Silence will not be treated as approval.

- **Responsible:** Database developer and interface designer; project leader coordinates.
- **Stakeholders:** IT Asset Management manager (approver), technicians and license administrators (reviewers), Information Security.
- **Deliverables:** Approved design document, data model, screen and report mock-ups, working dummy.
- **Decision point:** What will the result look like, and is it desirable?

Phase 4: Development

During the development phase, the team arranges everything needed for implementation so it knows exactly what must be done, by whom, and when (Baars, 2006). I will break the work into a work-breakdown structure and a Gantt chart covering table building, form and report creation, data migration, testing, and training. The system administrator will set up separate development, test, and production environments. The data specialist will collect existing departmental spreadsheets and purchasing records for migration. The QA tester will write a test plan linked directly to each requirement.

I will also confirm staff availability. The handbook describes a project that lost a key member to unplanned leave, resulting in additional costs for an outside specialist (Baars, 2006). To avoid similar issues, I will review vacation and academic calendars, schedule go-live dates outside peak IT support periods, and include buffer time between tasks. I will also coordinate with department supervisors to confirm each team member's project hours, as overcommitting staff is a common cause of delays (Baars, 2006, Appendix 1).

- **Responsible:** Project leader, system administrator, data specialist, QA tester.
- **Deliverables:** Action plan, detailed implementation schedule and budget, prepared environments, test plan.
- **Decision point:** How will we implement the project, and is it feasible?

Phase 5: Implementation

In the implementation phase, the actual result is built (Baars, 2006). Because the handbook notes that software requirements are rarely complete at the start, I will organize this phase into two-week work cycles, as in the DANS method, so the customer can review each piece as it is finished (Baars, 2006, Chapter 6). The developers will first build the core tables for

software and licenses, then installations and computers, then reports. The data specialist will clean and import the legacy spreadsheets, remove duplicates, and standardize vendor names.

Testing happens throughout. The QA tester will check each feature against the requirements list, and technicians will perform acceptance tests. Before campus-wide rollout, the database will be piloted with two units, the College of Engineering and the University Library, for four weeks. The pilot will measure data correctness (a sample of 100 computers compared with what is actually installed), how long it takes to enter a new installation, and how long it takes to produce a mock audit report. Weekly team meetings will use the handbook's issue log, risk log, and meeting report templates (Baars, 2006, Appendices 7–9). The team will discuss and register any changes to the approved requirements or design so decisions can be traced later (Baars, 2006).

- **Responsible:** Database developer and programmer (build), data specialist (migration), QA tester (testing), project leader (progress and change control).
- **Stakeholders:** Pilot departments, IT Asset Management, technicians.
- **Deliverables:** A working, tested database with migrated data; a pilot evaluation report; signed customer acceptance.
- **Decision point:** Is the result adequate, and how will we arrange evaluation?

Phase 6: Follow-up

Although it is often neglected, the follow-up phase brings the project to a proper close (Baars, 2006). The team will write a user manual and quick-reference guides, run applied training sessions for technicians and license administrators, and roll the database out to the

remaining colleges in stages. The team will formally transfer ownership to the IT Operations database administrator through a handover document.

The handbook notes that most follow-up problems stem from agreements that were never made (Baars, 2006). The handover will therefore answer its key questions: how long the project team provides support (four weeks after go-live), how quickly errors must be fixed, who handles updates and new report requests, and who owns and may change the database. Once these limits are reached, the project ends, and new requests go to IT Operations. I will then compile the final report and financial statement comparing budgeted and actual hours (Baars, 2006, Appendix 12), hold a lessons-learned session, thank the team, and dissolve the project team.

- **Responsible:** Project leader, business analyst (manuals and training), database administrator (receiving owner).
- **Deliverables:** User documentation, training, transfer documents, final project report, and financial statement.
- **Decision point:** Final report and formal project closure.

Table 1

Summary of Project Phases, Timeline, and Decision Points

1. Initiation	1–3	Approved project plan and global budget	Are we going to do this project?
2. Definition	4–7	Approved list of requirements (and exclusions)	Which requirements must be met? Is it possible?
3. Design	8–11	Approved design document and mock-ups	What will the result look like? Is it desirable?
4. Development	12–13	Action plan, schedule, ready environments	How will we build it? Is it feasible?
5. Implementation	14–23	Tested, piloted, and	Is the result adequate?

		accepted database	
6. Follow-up	24–26	Training, handover documents, final report	Final report; project closed

Note. Decision-point questions adapted from Baars (2006, Chapter 3).

Conclusion

Developing a software-tracking database for a university requires coordination among diverse groups, including auditors, legal counsel, and daily users. The six-phase model provides a structured approach: first establishing purpose and requirements, then determining implementation, and finally building the solution (Baars, 2006). Setting clear boundaries, documenting requirements and approvals, involving end users early, piloting before full rollout, and ensuring a defined handover address common causes of project failure. Decision points at each phase allow the team and CIO to adjust time, budget, and scope proactively, reducing the risk of late-stage issues.

References

Baars, W. (2006). *Project management handbook* (Version 1.1). DANS – Data Archiving and Networked Services. http://www.projectmanagement-training.net